Topic: Undetermined coefficients for nonhomogeneous systems

Question: Given the vector F from a system $\overrightarrow{x}' = A\overrightarrow{x} + F$, and without knowing the complementary solution of the system, what should be the initial guess for the particular solution?

$$F = \begin{bmatrix} \cos(2t) \\ -\sin t \\ t^2 - 3 \end{bmatrix}$$

Answer choices:

A $\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \cos(2t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \sin t + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} t^2$

B $\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \cos(2t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \sin t + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} t^2 + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

C $\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \sin(2t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \cos(2t) + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \sin t + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \cos t + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} t^2 + \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$

D $\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \sin(2t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \cos(2t) + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \sin t + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \cos t$

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$$+\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} t^2 + \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} t + \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}$$

Solution: D

Start by rewriting F as the sum of separate vectors.

$$F = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cos(2t) + \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \sin t + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} t^2 + \begin{bmatrix} 0 \\ 0 \\ -3 \end{bmatrix}$$

The guess for the particular solution for the $\cos(2t)$ portion will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \sin(2t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \cos(2t)$$

The guess for the particular solution for the $\sin t$ portion will be

$$\overrightarrow{x_p} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \sin t + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \cos t$$

The guess for the particular solution for the $t^2 - 3$ portion will be

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$$\overrightarrow{x_p} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} t^2 + \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} t + \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}$$

So the guess for the complete particular solution, without accounting for any overlap with the complementary solution, will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \sin(2t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \cos(2t) + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \sin t + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \cos t$$

$$+ \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} t^2 + \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} t + \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}$$

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Topic: Undetermined coefficients for nonhomogeneous systems

Question: Use the method of undetermined coefficients to find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix} \vec{x} + \begin{bmatrix} -e^{3t} \\ t^2 + 5 \end{bmatrix}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{5t} + \begin{bmatrix} -e^{3t} \\ \frac{1}{2}e^{3t} - \frac{1}{5}t^2 - \frac{2}{25}t - \frac{127}{125} \end{bmatrix}$$

B $$\vec{x} = c_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t} + \begin{bmatrix} -e^{3t} \\ \frac{1}{2}e^{3t} - \frac{1}{5}t^2 - \frac{2}{25}t - \frac{127}{125} \end{bmatrix}$$

C $$\vec{x} = c_1 \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{5t} + \begin{bmatrix} e^{3t} \\ 2e^{3t} + \frac{1}{5}t^2 - 2t + 1 \end{bmatrix}$$

D $$\vec{x} = c_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t} + \begin{bmatrix} e^{3t} \\ 2e^{3t} + \frac{1}{5}t^2 - 2t + 1 \end{bmatrix}$$

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Solution: B

We have to start by finding the complementary solution, which means we'll find $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 0 \\ 1 & 5 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (2 - \lambda)(5 - \lambda) - (0)(1)$$

$$|A - \lambda I| = (2 - \lambda)(5 - \lambda)$$

So the characteristic equation is

$$(2 - \lambda)(5 - \lambda) = 0$$

$$\lambda = 2, 5$$

Then for these Eigenvalues, $\lambda_1 = 2$ and $\lambda_2 = 5$, we find

$$A - 2I = \begin{bmatrix} 2 - 2 & 0 \\ 1 & 5 - 2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix}$$

and

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$$A - 5I = \begin{bmatrix} 2-5 & 0 \\ 1 & 5-5 \end{bmatrix}$$

$$A - 5I = \begin{bmatrix} -3 & 0 \\ 1 & 0 \end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ -3 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1 + 3k_2 = 0$$

$$k_1 = -3k_2$$

$$k_1 = 0$$

For the first system, we choose $(k_1, k_2) = (3, -1)$. And from the second system, we choose $(k_1, k_2) = (0,1)$.

$$\vec{k_1} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\vec{k_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Then the solutions to the system are

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t}$$

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

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$$\overrightarrow{x_1} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^{2t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t}$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

$$F = \begin{bmatrix} -e^{3t} \\ t^2 + 5 \end{bmatrix}$$

$$F = \begin{bmatrix} -1 \\ 0 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} t^2 + \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^{3t} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} t^2 + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} t + \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

$$\overrightarrow{x_p}' = \begin{bmatrix} 3a_1 \\ 3a_2 \end{bmatrix} e^{3t} + \begin{bmatrix} 2b_1 \\ 2b_2 \end{bmatrix} t + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Plugging this into the matrix equation representing the system of differential equations $\overrightarrow{x}' = A\overrightarrow{x} + F$, we get

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$$\begin{bmatrix}3a_1\\3a_2\end{bmatrix}e^{3t}+\begin{bmatrix}2b_1\\2b_2\end{bmatrix}t+\begin{bmatrix}c_1\\c_2\end{bmatrix}=\begin{bmatrix}2&0\\1&5\end{bmatrix}\left[\begin{bmatrix}a_1\\a_2\end{bmatrix}e^{3t}+\begin{bmatrix}b_1\\b_2\end{bmatrix}t^2+\begin{bmatrix}c_1\\c_2\end{bmatrix}t+\begin{bmatrix}d_1\\d_2\end{bmatrix}\right]$$

$$+\begin{bmatrix}-1\\0\end{bmatrix}e^{3t}+\begin{bmatrix}0\\1\end{bmatrix}t^2+\begin{bmatrix}0\\5\end{bmatrix}$$

$$\begin{bmatrix}3a_1e^{3t}+2b_1t+c_1\\3a_2e^{3t}+2b_2t+c_2\end{bmatrix}=\begin{bmatrix}2&0\\1&5\end{bmatrix}\begin{bmatrix}a_1e^{3t}+b_1t^2+c_1t+d_1\\a_2e^{3t}+b_2t^2+c_2t+d_2\end{bmatrix}+\begin{bmatrix}-e^{3t}\\t^2+5\end{bmatrix}$$

Breaking this equation into a system of two equations gives

$$3a_1e^{3t}+2b_1t+c_1=2(a_1e^{3t}+b_1t^2+c_1t+d_1)+0(a_2e^{3t}+b_2t^2+c_2t+d_2)-e^{3t}$$

$$3a_2e^{3t}+2b_2t+c_2=1(a_1e^{3t}+b_1t^2+c_1t+d_1)+5(a_2e^{3t}+b_2t^2+c_2t+d_2)+t^2+5$$

which simplifies to

$$3a_1e^{3t}+2b_1t+c_1=(2a_1-1)e^{3t}+2b_1t^2+2c_1t+2d_1$$

$$3a_2e^{3t}+2b_2t+c_2=(a_1+5a_2)e^{3t}+(b_1+5b_2+1)t^2+(c_1+5c_2)t+(d_1+5d_2+5)$$

These equations can each be broken into its own system.

$$3a_1=2a_1-1 \qquad 3a_2=a_1+5a_2$$

$$0=2b_1 \qquad 0=b_1+5b_2+1$$

$$2b_1=2c_1 \qquad 2b_2=c_1+5c_2$$

$$c_1=2d_1 \qquad c_2=d_1+5d_2+5$$

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Solving these eight equations as a system gives $(a_1, a_2) = (-1, 1/2)$, $(b_1, b_2) = (0, -1/5)$, $(c_1, c_2) = (0, -2/25)$, and $(d_1, d_2) = (0, -127/125)$. Therefore, the particular solution becomes

$$\overrightarrow{x_p} = \begin{bmatrix} -1 \\ \frac{1}{2} \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ -\frac{1}{5} \end{bmatrix} t^2 + \begin{bmatrix} 0 \\ -\frac{2}{25} \end{bmatrix} t + \begin{bmatrix} 0 \\ -\frac{127}{125} \end{bmatrix}$$

Now we can rewrite the particular solution as one vector.

$$\overrightarrow{x_p} = \begin{bmatrix} -e^{3t} \\ \frac{1}{2}e^{3t} - \frac{1}{5}t^2 - \frac{2}{25}t - \frac{127}{125} \end{bmatrix}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t} + \begin{bmatrix} -e^{3t} \\ \frac{1}{2}e^{3t} - \frac{1}{5}t^2 - \frac{2}{25}t - \frac{127}{125} \end{bmatrix}$$

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Topic: Undetermined coefficients for nonhomogeneous systems

Question: Use the method of undetermined coefficients to find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \vec{x} + \begin{bmatrix} t+2 \\ -3e^{3t} \\ \sin t \end{bmatrix}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} e^{(2+\sqrt{2})t} + c_3 \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} e^{(2-\sqrt{2})t} + \begin{bmatrix} -\frac{3}{4}t - \frac{19}{8} + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ -\frac{1}{4}t - \frac{9}{8} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ \frac{1}{2}t + 2 + 3e^{3t} - \frac{6}{17}\sin t - \frac{7}{17}\cos t \end{bmatrix}$$

B $$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} te^{-2t}$$

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$$+\begin{bmatrix} -\frac{3}{4}t - \frac{19}{8} + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ -\frac{1}{4}t - \frac{9}{8} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ \frac{1}{2}t + 2 + 3e^{3t} - \frac{6}{17}\sin t - \frac{7}{17}\cos t \end{bmatrix}$$

C $\quad \vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} e^{(2+\sqrt{2})t} + c_3 \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} e^{(2-\sqrt{2})t}$

$$+\begin{bmatrix} -6t - 19 + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ -2t - 9 + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ 4t + 16 + 3e^{3t} - \frac{6}{17}\sin t - \frac{7}{17}\cos t \end{bmatrix}$$

D $\quad \vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} te^{-2t}$

$$+\begin{bmatrix} -6t - 19 + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ -2t - 9 + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ 4t + 16 + 3e^{3t} - \frac{6}{17}\sin t - \frac{7}{17}\cos t \end{bmatrix}$$

Solution: A

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We have to start by finding the complementary solution, which means we'll find $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 0 & 1 \\ 0 & 2-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 \\ 1 & 2-\lambda \end{vmatrix} + 1 \begin{vmatrix} 0 & 2-\lambda \\ 1 & 1 \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda)[(2-\lambda)(2-\lambda) - (1)(1)] + [(0)(1) - (2-\lambda)(1)]$$

$$|A - \lambda I| = (2-\lambda)(3 - 4\lambda + \lambda^2) - (2-\lambda)$$

$$|A - \lambda I| = (2-\lambda)(3 - 4\lambda + \lambda^2 - 1)$$

$$|A - \lambda I| = (2-\lambda)(\lambda^2 - 4\lambda + 2)$$

So the characteristic equation is

$$(2-\lambda)(\lambda^2 - 4\lambda + 2) = 0$$

$$\lambda = 2,\ 2 \pm \sqrt{2}$$

Then for these Eigenvalues, $\lambda_1 = 2$, $\lambda_2 = 2 + \sqrt{2}$, and $\lambda_3 = 2 - \sqrt{2}$, we find

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$$A - 2I = \begin{bmatrix} 2-2 & 0 & 1 \\ 0 & 2-2 & 1 \\ 1 & 1 & 2-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

and

$$A - (2+\sqrt{2})I = \begin{bmatrix} 2-(2+\sqrt{2}) & 0 & 1 \\ 0 & 2-(2+\sqrt{2}) & 1 \\ 1 & 1 & 2-(2+\sqrt{2}) \end{bmatrix}$$

$$A - (2+\sqrt{2})I = \begin{bmatrix} -\sqrt{2} & 0 & 1 \\ 0 & -\sqrt{2} & 1 \\ 1 & 1 & -\sqrt{2} \end{bmatrix}$$

and

$$A - (2-\sqrt{2})I = \begin{bmatrix} 2-(2-\sqrt{2}) & 0 & 1 \\ 0 & 2-(2-\sqrt{2}) & 1 \\ 1 & 1 & 2-(2-\sqrt{2}) \end{bmatrix}$$

$$A - (2-\sqrt{2})I = \begin{bmatrix} \sqrt{2} & 0 & 1 \\ 0 & \sqrt{2} & 1 \\ 1 & 1 & \sqrt{2} \end{bmatrix}$$

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Put these three matrices into reduced row-echelon form.

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -\sqrt{2} & 0 & 1 \\ 0 & -\sqrt{2} & 1 \\ 1 & 1 & -\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{2} & 0 & 1 \\ 0 & \sqrt{2} & 1 \\ 1 & 1 & \sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -\sqrt{2} \\ -\sqrt{2} & 0 & 1 \\ 0 & -\sqrt{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & \sqrt{2} \\ \sqrt{2} & 0 & 1 \\ 0 & \sqrt{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -\sqrt{2} \\ 0 & \sqrt{2} & -1 \\ 0 & -\sqrt{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & \sqrt{2} \\ 0 & -\sqrt{2} & -1 \\ 0 & \sqrt{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -\sqrt{2} \\ 0 & 1 & -\frac{\sqrt{2}}{2} \\ 0 & -\sqrt{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & \sqrt{2} \\ 0 & 1 & \frac{\sqrt{2}}{2} \\ 0 & \sqrt{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & -\frac{\sqrt{2}}{2} \\ 0 & -\sqrt{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & \frac{\sqrt{2}}{2} \\ 0 & \sqrt{2} & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1 + k_2 = 0 \qquad k_1 - \frac{\sqrt{2}}{2}k_3 = 0 \qquad k_1 + \frac{\sqrt{2}}{2}k_3 = 0$$

$$k_3 = 0 \qquad k_2 - \frac{\sqrt{2}}{2}k_3 = 0 \qquad k_2 + \frac{\sqrt{2}}{2}k_3 = 0$$

Rewriting these systems gives

$$k_1 = -k_2 \qquad k_1 = \frac{\sqrt{2}}{2}k_3 \qquad k_1 = -\frac{\sqrt{2}}{2}k_3$$

$$k_3 = 0 \qquad k_2 = \frac{\sqrt{2}}{2}k_3 \qquad k_2 = -\frac{\sqrt{2}}{2}k_3$$

For the first system, we choose $(k_1, k_2, k_3) = (1, -1, 0)$, from the second system, we choose $(k_1, k_2, k_3) = (1, 1, \sqrt{2})$, and from the third system, we choose $(k_1, k_2, k_3) = (1, -1, \sqrt{2})$.

$$\vec{k_1} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \qquad \vec{k_2} = \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} \qquad \vec{k_3} = \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix}$$

Then the solutions to the system are

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$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t} \qquad \overrightarrow{x_3} = \overrightarrow{k_3}e^{\lambda_3 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{2t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} e^{(2+\sqrt{2})t} \qquad \overrightarrow{x_3} = \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} e^{(2-\sqrt{2})t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} e^{(2+\sqrt{2})t} + c_3 \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} e^{(2-\sqrt{2})t}$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

$$F = \begin{bmatrix} t+2 \\ -3e^{3t} \\ \sin t \end{bmatrix}$$

$$F = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} t + \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \sin t$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} t + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} e^{3t} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \sin t + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} \cos t$$

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$$\overrightarrow{x_p}' = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + \begin{bmatrix} 3c_1 \\ 3c_2 \\ 3c_3 \end{bmatrix} e^{3t} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \cos t + \begin{bmatrix} -e_1 \\ -e_2 \\ -e_3 \end{bmatrix} \sin t$$

Plugging this into the matrix equation representing the system of differential equations $\overrightarrow{x}' = A\overrightarrow{x} + F$, we get

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + \begin{bmatrix} 3c_1 \\ 3c_2 \\ 3c_3 \end{bmatrix} e^{3t} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \cos t + \begin{bmatrix} -e_1 \\ -e_2 \\ -e_3 \end{bmatrix} \sin t$$

$$= \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \left[\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} t + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} e^{3t} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \sin t + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} \cos t \right]$$

$$+ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} t + \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \sin t$$

$$\begin{bmatrix} a_1 + 3c_1e^{3t} + d_1 \cos t - e_1 \sin t \\ a_2 + 3c_2e^{3t} + d_2 \cos t - e_2 \sin t \\ a_3 + 3c_3e^{3t} + d_3 \cos t - e_3 \sin t \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} a_1t + b_1 + c_1e^{3t} + d_1 \sin t + e_1 \cos t \\ a_2t + b_2 + c_2e^{3t} + d_2 \sin t + e_2 \cos t \\ a_3t + b_3 + c_3e^{3t} + d_3 \sin t + e_3 \cos t \end{bmatrix} + \begin{bmatrix} t + 2 \\ -3e^{3t} \\ \sin t \end{bmatrix}$$

Breaking this equation into a system of three equations gives

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$$a_1 + 3c_1e^{3t} + d_1\cos t - e_1\sin t = 2(a_1t + b_1 + c_1e^{3t} + d_1\sin t + e_1\cos t)$$
$$+0(a_2t + b_2 + c_2e^{3t} + d_2\sin t + e_2\cos t)$$
$$+1(a_3t + b_3 + c_3e^{3t} + d_3\sin t + e_3\cos t) + t + 2$$

$$a_2 + 3c_2e^{3t} + d_2\cos t - e_2\sin t = 0(a_1t + b_1 + c_1e^{3t} + d_1\sin t + e_1\cos t)$$
$$+2(a_2t + b_2 + c_2e^{3t} + d_2\sin t + e_2\cos t)$$
$$+1(a_3t + b_3 + c_3e^{3t} + d_3\sin t + e_3\cos t) - 3e^{3t}$$

$$a_3 + 3c_3e^{3t} + d_3\cos t - e_3\sin t = 1(a_1t + b_1 + c_1e^{3t} + d_1\sin t + e_1\cos t)$$
$$+1(a_2t + b_2 + c_2e^{3t} + d_2\sin t + e_2\cos t)$$
$$+2(a_3t + b_3 + c_3e^{3t} + d_3\sin t + e_3\cos t) + \sin t$$

which simplifies to

$$a_1 + 3c_1e^{3t} + d_1\cos t - e_1\sin t = 2a_1t + 2b_1 + 2c_1e^{3t} + 2d_1\sin t + 2e_1\cos t$$
$$+a_3t + b_3 + c_3e^{3t} + d_3\sin t + e_3\cos t + t + 2$$

$$a_2 + 3c_2e^{3t} + d_2\cos t - e_2\sin t = 2a_2t + 2b_2 + 2c_2e^{3t} + 2d_2\sin t + 2e_2\cos t$$
$$+a_3t + b_3 + c_3e^{3t} + d_3\sin t + e_3\cos t - 3e^{3t}$$

$$a_3 + 3c_3e^{3t} + d_3\cos t - e_3\sin t = a_1t + b_1 + c_1e^{3t} + d_1\sin t + e_1\cos t$$
$$+a_2t + b_2 + c_2e^{3t} + d_2\sin t + e_2\cos t$$
$$+2a_3t + 2b_3 + 2c_3e^{3t} + 2d_3\sin t + 2e_3\cos t + \sin t$$

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and then to

$$(a_1) + (3c_1)e^{3t} + (d_1)\cos t + (-e_1)\sin t = (2a_1 + a_3 + 1)t + (2b_1 + b_3 + 2)$$

$$+(2c_1 + c_3)e^{3t} + (2e_1 + e_3)\cos t + (2d_1 + d_3)\sin t$$

$$(a_2) + (3c_2)e^{3t} + (d_2)\cos t + (-e_2)\sin t = (2a_2 + a_3)t + (2b_2 + b_3)$$

$$+(2c_2 + c_3 - 3)e^{3t} + (2e_2 + e_3)\cos t + (2d_2 + d_3)\sin t$$

$$(a_3) + (3c_3)e^{3t} + (d_3)\cos t + (-e_3)\sin t = (a_1 + a_2 + 2a_3)t + (b_1 + b_2 + 2b_3)$$

$$+(c_1 + c_2 + 2c_3)e^{3t} + (e_1 + e_2 + 2e_3)\cos t + (d_1 + d_2 + 2d_3 + 1)\sin t$$

These equations can each be broken into its own system.

$$0 = 2a_1 + a_3 + 1 \qquad 0 = 2a_2 + a_3 \qquad 0 = a_1 + a_2 + 2a_3$$

$$a_1 = 2b_1 + b_3 + 2 \qquad a_2 = 2b_2 + b_3 \qquad a_3 = b_1 + b_2 + 2b_3$$

$$3c_1 = 2c_1 + c_3 \qquad 3c_2 = 2c_2 + c_3 - 3 \qquad 3c_3 = c_1 + c_2 + 2c_3$$

$$d_1 = 2e_1 + e_3 \qquad d_2 = 2e_2 + e_3 \qquad d_3 = e_1 + e_2 + 2e_3$$

$$-e_1 = 2d_1 + d_3 \qquad -e_2 = 2d_2 + d_3 \qquad -e_3 = d_1 + d_2 + 2d_3 + 1$$

Solving these 15 equations as a system gives $(a_1, a_2, a_3) = (-3/4, -1/4, 1/2)$, $(b_1, b_2, b_3) = (-19/8, -9/8, 2)$, $(c_1, c_2, c_3) = (3, 0, 3)$, $(d_1, d_2, d_3) = (1/17, 1/17, -6/17)$, and $(e_1, e_2, e_3) = (4/17, 4/17, -7/17)$. Therefore, the particular solution becomes

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$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{3}{4} \\ -\frac{1}{4} \\ \frac{1}{2} \end{bmatrix} t + \begin{bmatrix} -\frac{19}{8} \\ -\frac{9}{8} \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} e^{3t} + \begin{bmatrix} \frac{1}{17} \\ \frac{1}{17} \\ -\frac{6}{17} \end{bmatrix} \sin t + \begin{bmatrix} \frac{4}{17} \\ \frac{4}{17} \\ -\frac{7}{17} \end{bmatrix} \cos t$$

Now we can rewrite the particular solution as one vector.

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{3}{4}t - \frac{19}{8} + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ -\frac{1}{4}t - \frac{9}{8} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ \frac{1}{2}t + 2 + 3e^{3t} - \frac{6}{17}\sin t - \frac{7}{17}\cos t \end{bmatrix}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} e^{(2+\sqrt{2})t} + c_3 \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} e^{(2-\sqrt{2})t}$$

$$+ \begin{bmatrix} -\frac{3}{4}t - \frac{19}{8} + 3e^{3t} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ -\frac{1}{4}t - \frac{9}{8} + \frac{1}{17}\sin t + \frac{4}{17}\cos t \\ \frac{1}{2}t + 2 + 3e^{3t} - \frac{6}{17}\sin t - \frac{7}{17}\cos t \end{bmatrix}$$

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